

Worksheet W1: The Big Secret

"Calculus is just arithmetic for things that move."

Chapter 1 · 10 Problems · Name: _____ Date: _____

Problem 1

Question: What are the two questions that all of calculus answers?

💡 *Hint: Think about a speedometer and an odometer.*

Problem 2

Question: What is the difference between **average speed** and **instant speed**?

💡 *Hint: Average speed is total distance $\div$ total time. Instant speed is at a single moment.*

Problem 3

Question: If a car travels 120 miles in 2 hours, what is its **average speed**?

💡 *Hint: Average speed = Total distance $\div$ Total time.*

Problem 4

Question: The **derivative** is like a _____, and the **integral** is like an _____.

💡 *Hint: One tells you speed right now. The other tells you total distance.*

Problem 5

Question: Why is $\frac{0}{0}$ a problem when calculating instant speed?

💡 *Hint: At an instant, what are the values of distance and time?*

Problem 6

Question: Complete this sentence: "Calculus is just arithmetic for _____."

💡 *Hint: Think about the "Big Secret" from Chapter 1.*

Problem 7

Question: What does the "**Mental Zoom**" help us visualize?

💡 *Hint: When you zoom in on a curve, what does it look like?*

Problem 8

Question: If a car drives at 60 mph for 3 hours, how far does it travel?

💡 *Hint: Distance = Speed $\times$ Time.*

Problem 9

Question: What is the **limit** trick used for in calculus?

💡 *Hint: When you can't divide 0 by 0, you look at what the average speed approaches.*

Problem 10

Question: A car's distance is given by $d = 2t^2$. What is the instant speed at $t = 2$ seconds?
(Use: speed = $4t$)

💡 *Hint: Plug $t = 2$ into $4t$.*